

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 4: Revealed Preference and Afriat's Theorem

Notes on the Chapter

Chapter 4 provides a statement and proof of Afriat's Theorem and then applies the theorem to show that, in theory, Giffen goods are possible. Except for the proof of Afriat's Theorem (I use the Varian's proof), everything is pretty direct and straightforward, and while it may be worthwhile to go over an example illustrating how demand data is used to infer revealed weak and strict preference, most students seem to get the basic points of the chapter fairly readily. As I noted in the discussion of Chapter 3, I rarely have spent more than half a session (a full session being 90 to 105 minutes) on this chapter; since Chapter 3 takes me a session and a half, the two fit well together.

I do make a point of emphasizing that GARP (the generalized axiom of revealed preference, or no strict preference cycles found in the data) is necessary for the model of a utility-maximizing consumer (with locally insatiable preferences) to apply, but is sufficient for the construction of a consumer whose utility function is continuous, strictly increasing, and convex. Hence those additional properties generate no testable restrictions for a finite amount of demand data.

The chapter does not discuss the classic literature in which revealed preference is applied to demand functions. That discussion—such that it is—happens in Chapter 11.

Notes on the Problems

None of the problems in this chapter should present a challenge to students.

Solutions to Non-Starred Problems

- 4.2. Suppose that f_1 through f_n are (finitely many) real-valued, concave functions defined on a common convex domain in R^k . Take any x and x' from the common

domain of these functions, and any $\alpha \in [0, 1]$. We know, then, that $f_i(\alpha x + (1 - \alpha)x') \geq \alpha f_i(x) + (1 - \alpha)f_i(x')$ for each $i = 1, \dots, n$. Therefore,

$$\min_{i=1, \dots, n} \{f_i(\alpha x + (1 - \alpha)x')\} \geq \min_{i=1, \dots, n} \{\alpha f_i(x) + (1 - \alpha)f_i(x')\}.$$

I assert, moreover, that

$$\min_{i=1, \dots, n} \{\alpha f_i(x) + (1 - \alpha)f_i(x')\} \geq \alpha \min_{i=1, \dots, n} \{f_i(x)\} + (1 - \alpha) \min_{i=1, \dots, n} \{f_i(x')\},$$

since if j is the minimizing index on the left-hand side, the two minima on the right-hand side are no larger than $f_j(x)$ and, respectively, $f_j(x')$. But then if $g = \min_i f_i$, these two inequalities run together say that

$$g(\alpha x + (1 - \alpha)x') \geq \alpha g(x) + (1 - \alpha)g(x'),$$

that is, g is concave.

(If a finite list of functions f_1 through f_n defined on a common convex domain are all quasi-concave, is $g = \min_i f_i$ also quasi-concave? It is; can you find a simple proof?)

Next suppose that the functions f_1 through f_n are strictly increasing on their (common) domains (which need no longer be convex sets), and suppose x and x' are two points from the common domain with $x \geq x'$ and $x \neq x'$. We know that $f_i(x') < f_i(x)$ for all i . Let $g = \min_i f_i$, and suppose j is the index which achieves this minimum at x . That is $g(x) = f_j(x)$. Then

$$g(x') = \min_i f_i(x') \leq f_j(x') < f_j(x) = g(x),$$

and since x and x' here are arbitrarily chosen points with $x \geq x'$ and $x \neq x'$, this shows that g is strictly increasing.

Suppose that the functions f_1 through f_n are continuous on their domains. Let x be any point in that (common) domain, and let $\{x^m\}$ be any sequence of points from the domain that has limit x . Fix some $\epsilon > 0$. We know, since the functions f_i are all continuous, that for some sufficiently large M and for all $m \geq M$ and $i = 1, \dots, N$, $|f_i(x^m) - f_i(x)| \leq \epsilon$. Now let $g = \min_i f_i$, and let j be the index that achieves the minimum at x ; that is, $g(x) = f_j(x)$. I assert that for all $m \geq M$, $|g(x^m) - g(x)| \leq \epsilon$. We know on the one hand that $g(x^m) \leq f_j(x^m)$, so $g(x^m) - g(x) \leq f_j(x^m) - f_j(x) \leq \epsilon$. And if $g(x^m) = f_k(x^m)$ (where the specific index k can depend on x^m), then $g(x) - g(x^m) \leq f_k(x) - f_k(x^m)$ (since $g(x) \leq f_k(x)$) $\leq \epsilon$. That does it.

(Challenge: Which if any of these three results survive if we are looking at an infinite collection of functions f_α over which we take a pointwise infimum?)

Finally, in the proof of Lemma 4.5, if $I = J$, then $n(i) = 0$ for all the x^i , which means that $p^j \cdot x^i \geq y^i$ for all i and j , including $i = j$. Therefore, $p^j \cdot x^i \geq p^j \cdot x^j$ for all i and j . Let $\alpha^j = 1$ and $v^j = 0$ for all j , and (4.1) follows immediately.

■ 4.3. Here's a picture:

